Data Science Program, George Washington University

Sample Midterm Exam

DATS 6101 (Introduction to Data Science)

Date: October 26, 2026

Instructor: Prof. Sushovan Majhi

Name: _____

Please read the following instructions carefully.

- This is a **sample paper** for practice only, at the real length, the real shape and the real difficulty, covering everything through Oct 19.
- It prints **30 problems. Answer any 25.** Each is worth **6 points**, so the paper is marked out of **150**, and if you answer more than 25 **your best 25 count**.
- You have **2 hours 30 minutes**, the full session.
- Most problems are multiple choice and **exactly one option is correct**. Mark it clearly (☑). The rest say what they want.
- The exam is **open-book** and **open-note**, but **printed material only**: the chapters covered, the labs whose solutions have been released, the homework covered, and your own notes.
- **Laptops**, **tablets**, **phones** and **watches** must be switched off and put away. No AI assistance and no communication of any kind.
- Nothing has to be computed that cannot be done on paper: every critical value a question needs is **printed in the question**. A correct number with no sentence around it earns part marks.
- All notation has its usual meaning. You may use standard results directly by stating them appropriately. **Read the instructions for individual questions carefully** before answering.

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Question	Points	Score
1	6	
2	6	
3	6	
4	6	
5	6	
6	6	
7	6	
8	6	
9	6	
10	6	
11	6	
12	6	
13	6	
14	6	
15	6	
16	6	
17	6	
18	6	
19	6	
20	6	
21	6	
22	6	
23	6	
24	6	
25	6	
26	6	
27	6	
28	6	
29	6	
30	6	
Total:	180	

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Part A Reading a claim

1. (6 points) A study reports mean satisfaction **7.2 (95% CI 6.9 to 7.5)**. Which statement is defensible?

   - □ 95% of respondents rated between 6.9 and 7.5
   - □ There is a 95% probability the true mean is between 6.9 and 7.5
   - √ **If the study were repeated many times, about 95% of intervals built this way would contain the true mean**
   - □ The next respondent will rate between 6.9 and 7.5 with 95% probability

2. (6 points) Exam scores have mean 72 and standard deviation 11, from $n = 400$ students, so the standard error is 0.55. A report says *"almost every student scored within about half a point of 72."* The error is

   - √ **describing students with a number that describes the average of 400 of them**
   - □ using $n = 400$ when the class is larger
   - □ using the standard error instead of the variance
   - □ forgetting to multiply by 1.96

3. (6 points) A comparison returns $p = 0.30$. The correct reading is

   - □ there is a 30% chance the null hypothesis is true
   - □ the two groups are equal
   - √ **data at least this extreme are unremarkable if the null is true, which leaves both "no effect" and "too small a study" alive**
   - □ the study has 30% power

4. (6 points) An app's rating is estimated from **40,000** reviews, all left voluntarily. The interval is extremely narrow. This tells you

   - □ the rating is accurate to within the interval
   - □ the sample is large enough for the CLT, so the estimate is unbiased
   - □ nothing, because ratings are ordinal
   - √ **the rating is precise, and says nothing about whether reviewers resemble users**

5. (6 points) You test 10,000 simulated datasets in which the null is *exactly* true, and count how many give $p < 0.05$. You should see about

   - □ 0
   - □ 50
   - √ **500**
   - □ 9,500

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Part B Probability and random variables

6. (6 points) A condition affects **2%** of a population. A test detects it in 95% of those who have it and is negative in 90% of those who do not. Someone tests positive. The probability they have the condition is closest to

   □ 2%

   √ **16%**

   □ 50%

   □ 95%

7. (6 points) X has variance 9. Then $\mathrm{Var}(3X - 2)$ is

   □ 25

   □ 27

   □ 79

   √ **81**

8. (6 points) Which is a valid probability density on $[0, 3]$?

   √ $f(x) = 2x/9$

   □ $f(x) = 1/2$

   □ $f(x) = 3 - x$

   □ $f(x) = x/3$

9. (6 points) Of 500 site visits, 200 came from search and 300 from social; 40 of the search visits and 60 of the social visits converted. Converting and arriving from search are

   □ dependent, because social produced more conversions

   □ dependent, because the groups are different sizes

   √ **independent—the conversion rate is 20% in both groups**

   □ impossible to judge without a significance test

10. (6 points) Two fair dice are rolled. The events *"the first shows a six"* and *"the total is seven"* are

    □ dependent, because both involve the first die

    √ **independent, because the total is equally likely to be 7 whatever the first die shows**

    □ dependent, because a six makes a large total more likely

    □ mutually exclusive

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11. (6 points) **Short answer.** A delivery fee is \$0 with probability 0.5, \$5 with probability 0.3 and \$20 with probability 0.2. Compute $\mathrm{E}[X]$ and $\mathrm{Var}(X)$, showing the working.

    Answer. $\mathrm{E}[X] = 0(0.5) + 5(0.3) + 20(0.2) = 1.5 + 4 = \mathbf{5.5}$. $\mathrm{E}[X^2] = 25(0.3) + 400(0.2) = 7.5 + 80 = 87.5$, *so* $\mathrm{Var}(X) = 87.5 - 5.5^2 = 87.5 - 30.25 = \mathbf{57.25}$ *(standard deviation about \$7.57). Full marks need* $\mathrm{E}[X^2]$ *shown; quoting a variance with no working earns half.*

Part C Expectation, variance and the CLT

12. (6 points) An interval is currently ± 2.0 at 95% confidence. You want ± 1.0 at 90%, with σ unchanged. Then n must grow by a factor of about

$z_{0.975}$	1.960	$z_{0.95}$	1.645

    - □ 2
    - √ **2.8**
    - □ 3.4
    - □ 4

13. (6 points) A clustered survey of **800** has a true standard error **twice** what $s/\sqrt{800}$ reports. Its effective sample size is about

    - √ **200**
    - □ 400
    - □ 800
    - □ 1,600

14. (6 points) The central limit theorem says nothing about which of these?

    - □ the sampling distribution of $\bar{X}$ at large n
    - □ the mean of the sampling distribution of $\bar{X}$
    - □ the spread of the sampling distribution of $\bar{X}$
    - √ **the sampling distribution of the sample *median***

15. (6 points) The population is exponential, whose skewness is 2. You average samples of $n = 25$. The skewness of the sample means is closest to

    - □ 0
    - √ **0.4**
    - □ 1
    - □ 2

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16. (6 points) For a Cauchy population, the mean of 1,000 draws is no more precise than the mean of 10. This is because

    □ the draws are dependent

    □ the Cauchy is discrete

    □ 1,000 is still too small

    √ **the Cauchy has no finite variance, so $\sigma/\sqrt{n}$ has nothing to describe**

17. (6 points) Skewness of the sampling distribution of $\bar{X}$ falls like $1/\sqrt{n}$. To cut it to **one third**, you must multiply the sample size by

    □ 3

    √ **9**

    □ 27

    □ 81

Part D Confidence intervals

18. (6 points) Drawing $n = 12$ from a strongly right-skewed population, a 95% t interval covers the true mean about 88% of the time. Which change most directly fixes the shortfall?

    √ **a larger sample, so the sampling distribution of the mean is closer to Normal**

    □ dividing by n rather than $n - 1$ when computing s

    □ `qnorm(0.975)` in place of `qt(0.975, 11)`

    □ a larger multiplier from the t table

19. (6 points) **Short answer.** A sample of **100** tickets has $\bar{x} = 52$ and $s = 20$ minutes. Using $t_{0.975,99} = 1.984$, give the standard error and the 95% interval, with the working.

Answer. SE $= 20/\sqrt{100} = 2.0$*; half-width* $= 1.984 \times 2 = 3.97$*; interval* 52 ± 4.0*, that is* ***48.0 to 56.0****. The division by* $\sqrt{100}$ *must be visible; a bare interval earns half.*

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20. (6 points) A paper reports a mean of 12.4 with a 95% CI of 11.6 to 13.2 and a standard deviation of 4.0, but never gives the sample size. It is closest to

    □ 25

    √ **96**

    □ 250

    □ 400

21. (6 points) Reporting a **90%** interval instead of a 95% one, everything else fixed, multiplies the width by about

    √ **0.84**

    □ 0.90

    □ 0.95

    □ 1.19

22. (6 points) A trial converts **7 of 250**. `prop.test(7, 250, correct = FALSE)` returns an interval of 0.0136 to 0.0567 around $\hat{p} = 0.028$. It is **not** symmetric about $\hat{p}$ because

    □ of rounding in the endpoints

    □ `prop.test` applies a continuity correction

    □ the sample size is too small for symmetry

    √ **a rate near zero has more room to be larger than smaller, and the Wilson interval reflects that**

23. (6 points) Group A has mean 20.0 and group B 24.0, each with standard error 1.5. Using 1.96 and $\mathrm{SE}_{\mathrm{diff}} = \sqrt{1.5^2 + 1.5^2} = 2.12$, the 95% interval for the difference is

    □ 1.06 to 6.94, which excludes zero

    □ not computable, because the intervals overlap

    □ 0.94 to 7.06, which excludes zero

    √ -0.16 **to 8.16, which includes zero**

24. (6 points) **Short answer.** An analyst's 95% interval for mean session length narrows pleasingly once they exclude every session longer than an hour. In one or two sentences, say what they have actually reported.

    Answer. *An interval for the mean of a **different** population—sessions under an hour—not for the one they were asked about. It is narrower because the truncated sample is less variable, and its centre has moved down. Nothing in the output records that the question changed, which is what makes it dangerous rather than merely wrong. Credit for naming both the changed estimand and the direction of the shift.*

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Part E Testing

25. (6 points) Thirty agents are timed on the same task before and after a training course. To ask whether the mean time changed, use

    - □ a two-sample (Welch) t-test
    - □ a one-sample t-test on the times recorded afterwards
    - √ **a paired t-test**
    - □ a chi-squared test

26. (6 points) You want to know whether complaint type (four categories) is related to region (three regions). The test is

    - √ **a chi-squared test of independence on the 4×3 table**
    - □ `prop.test` on the four complaint types
    - □ a two-sample t-test on each pair of regions
    - □ one-way ANOVA across the three regions

27. (6 points) An experiment on 2,000 sessions per arm returns $p = 0.002$ and a 95% confidence interval for the lift of **0.1% to 0.4%**. The right summary is

    - □ a 99.8% probability the new arm is better
    - □ a large, clearly valuable improvement
    - √ **a difference that is hard to attribute to chance and too small to be worth much—significance and importance are different questions**
    - □ an inconclusive result, since the interval is narrow

28. (6 points) **Short answer.** A colleague reports *"no significant difference ($p = 0.6$, $n = 18$ per group)"* and proposes dropping the feature. In two sentences, say what is wrong with the inference and what you would ask for.

    Answer. *A large p-value with 18 per group is equally consistent with no effect and with an effect the study had almost no chance of detecting—absence of evidence, not evidence of absence. Ask for the* ***confidence interval for the difference****: if it runs from clearly-worse to clearly-better, the experiment simply did not answer the question, and a power calculation says how many users would. Credit for naming power or the interval; full marks for both.*

Part F Experiments

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29. (6 points) Randomly assigning users to two versions buys you

    √ **groups alike in expectation on everything, measured or not, so a difference in outcome can be attributed to the version**

    □ freedom from the need for a confidence interval

    □ identical groups

    □ a larger effective sample size

30. (6 points) **Short answer.** In a randomized trial, 30% of the treatment arm stops using the product before the outcome is measured, against 10% of the control arm. In one or two sentences, say why the randomization no longer protects the comparison.

    Answer. *This is* ***differential attrition****. Randomization made the arms comparable* at assignment*; the comparison at the end is between the 70% who stayed in treatment and the 90% who stayed in control, and those are no longer randomly formed groups. If the people who left were the ones doing badly, the treatment arm now looks better for the wrong reason. Credit for naming that what survives randomization is the assignment, not everything after it.*

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Thirty problems printed; your best twenty-five count.
A number with no working earns part marks; the sentence is what is being graded.